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Integrated Hotel Booking And Guest Experience Management System

Supervisor:

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CSA4305– Internet Programming

Date : 25-11-2025

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AGENDA

- Introduction
- Problem Statement
- Literature Review
- Architecture
- Techniques used
- Project Demo
- Result & Analysis
- Conclusion
- Future Work
- References

Introduction:



- Manual hotel booking involves phone calls, paperwork, and delays.
- Guests face difficulty checking room availability in real-time.



- An online reservation system simplifies the booking process.
- Allows users to browse rooms, check availability, and book instantly.



- Reduces administrative load for hotel staff.
- Enhances guest experience through automation & web-based access.

Problem Statement:



- Traditional hotel booking is time-consuming and involves manual effort.
- Lack of a centralized system for real-time room availability.



- Manual record-keeping increases the chances of double booking.
- Data inconsistency occurs due to outdated or incorrect booking logs.



- Guests do not have an easy, self-service online platform to make bookings.
- Hotels require a secure, efficient, user-friendly online reservation portal

Literature Review :

S.NO	DETAILS OF THE JOURNAL	ALGORITHM / METHODOLOGY USED	ADVANTAGES	DIS ADVANTAGES
1	Basheer et al. (2025). Online Hotel Reservations & Guest Satisfaction in Kashmir.	Survey analysis; UX evaluation	Provides user satisfaction insights; helps improve UI/UX	Region-specific; not technical
2	Nguyen (2025). Web-based Hotel Booking System.	PHP–MySQL; CRUD operations; session login	Simple, scalable; reduces manual work	Limited automation; basic features
3	Gunjal (2025). <i>Hotel Booking App using MERN.</i>	MERN stack; REST APIs; JWT auth	Fast, modern UI; scalable system	Complex setup; requires higher skills



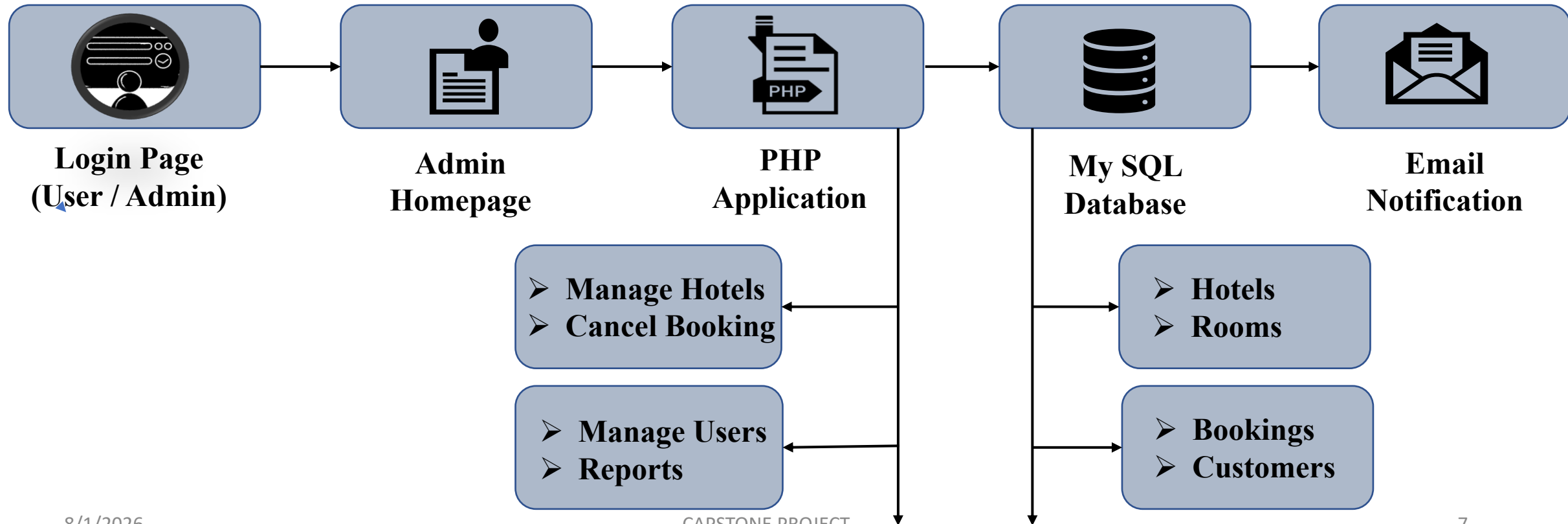
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MODULE 1 :Admin Panel (Hotel and Booking Control)

Architecture : (Admin side)





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TECHNIQUES USED

PHP

- Handles Server side Logic for hotels, rooms, and booking management
- Ensures structured navigation and modular admin workflows

CRUD Operations

- Admin can Create, Read, Update, Delete entries in all tables
- Ensures data integrity and real-time updates in the system

Admin Authentication

- Secure login maintains admin session across pages
- Prevents unauthorized access to admin-only operations



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Table Wise Data Management (MYSQL)

- Separate tables for hotels, rooms, bookings, customers
- Ensures organized storage and easy relational queries

Backend Validation & Error Handling

- Prevents duplicate entries, missing fields, or invalid operations
- Maintains cleanliness and accuracy of the database





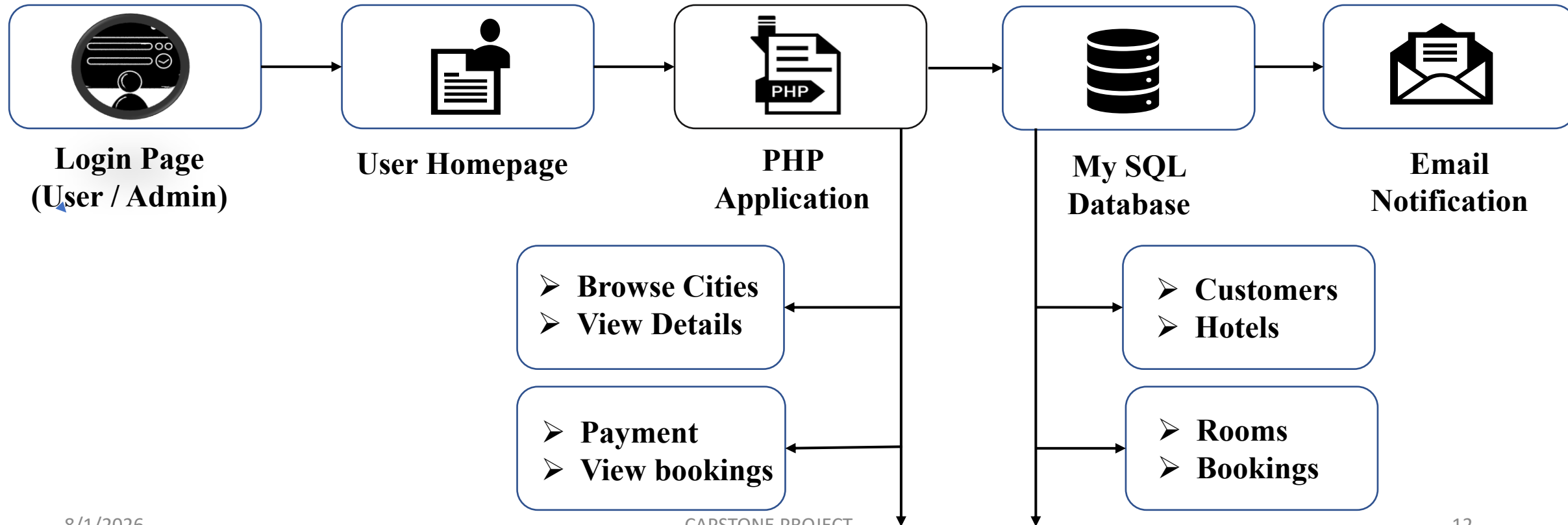
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MODULE 2 : User Portal (Booking & Guest Experience)

Architecture : (User Side)





PHP

- Displays cities, hotels, rooms Dynamically from the database
- Handles booking form submission and stores bookings securely

MYSQL for database Storage

- Stores user Account, hotels details, room data, and bookings.
- Ensures smooth retrieval using optimized SELECT queries.

HTML/CSS/Bootstrap

- Provides responsive, user-friendly layout on all devices.
- Enhances user experience with clean components like cards and forms.



Sessions for authentication

- Keeps the user Logged in across pages like Home, Book Room, My bookings.
- Ensures Personalized experience and secure access

SQL queries for data retrieval & booking logic

- Fetches hotels based on selected city, room availability, and pricing.
- Prevents double booking by checking room status before confirming.

TECHNIQUES

Frontend

HTML5, CSS3,
Bootstrap5

Responsive Card
Layouts for
hotels & rooms

Dynamic Hero
images and
Room Galleries

Backend

Core PHP for
Business Logic,
MYSQL

SQL JOIN
Queries for
room hotel
mapping

Session Based
Login System

Server

XAMPP (Apache
+ MYSQL)

Browser based
UI testing

Infinity Free for
Deployment



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MODULE-3 Payment & Billing System



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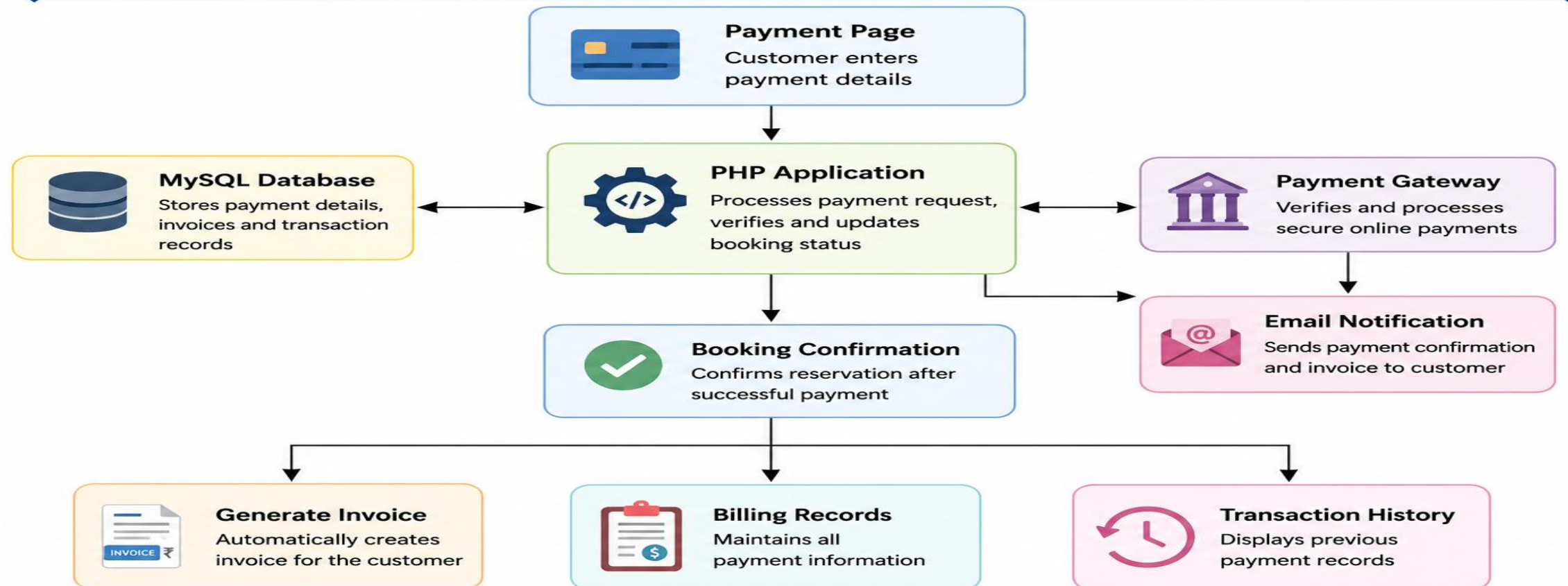


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PROJECT DEMO

MODULE 3 : PAYMENT & BILLING SYSTEM

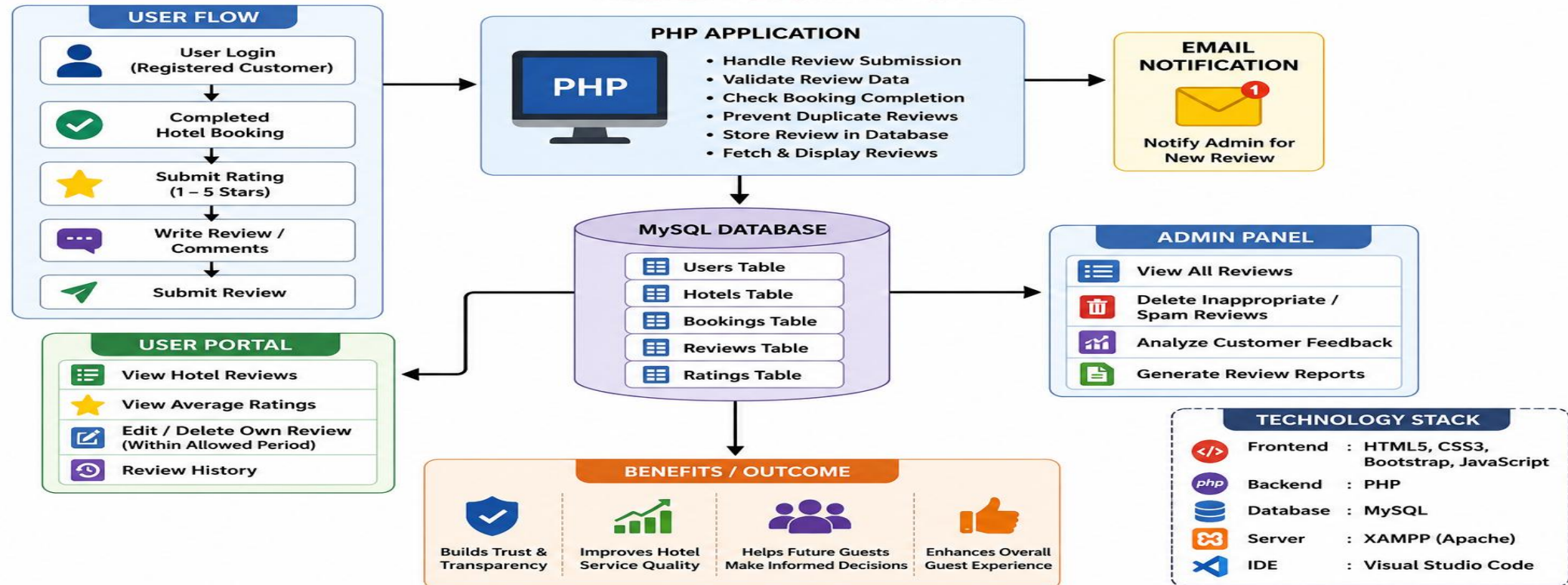
Architecture : (Payment & Billing System)



MODULE-4 Review system

MODULE-4 : FEEDBACK & REVIEW MANAGEMENT

ARCHITECTURE DIAGRAM





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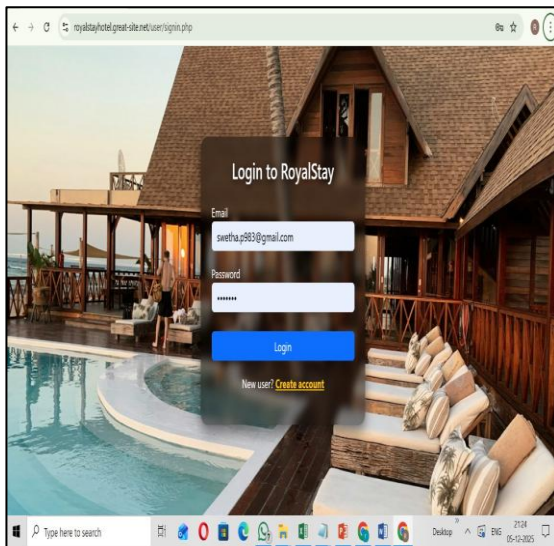


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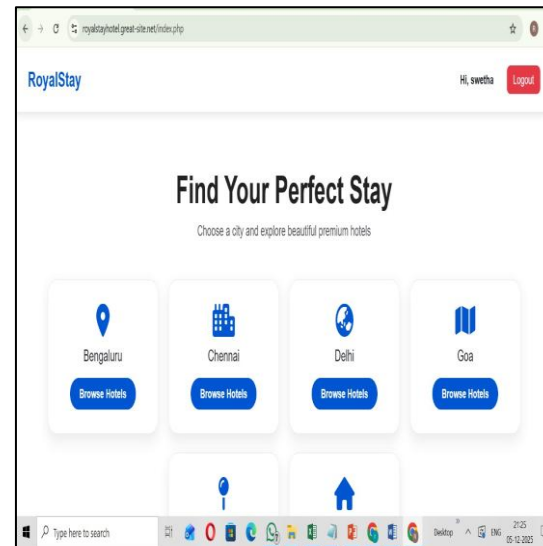
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Admin and User Interface

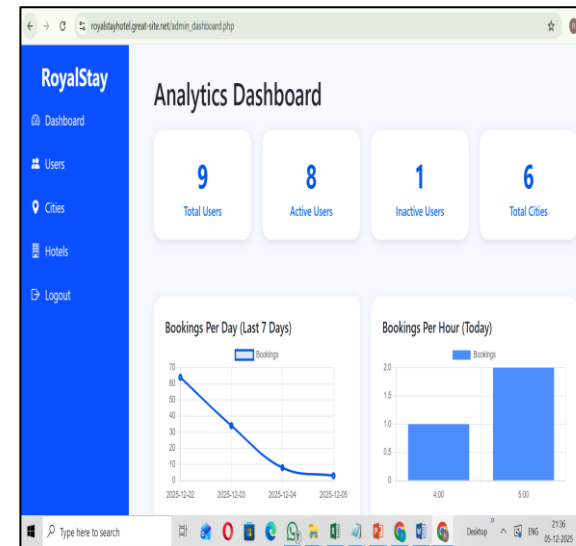
Authentication



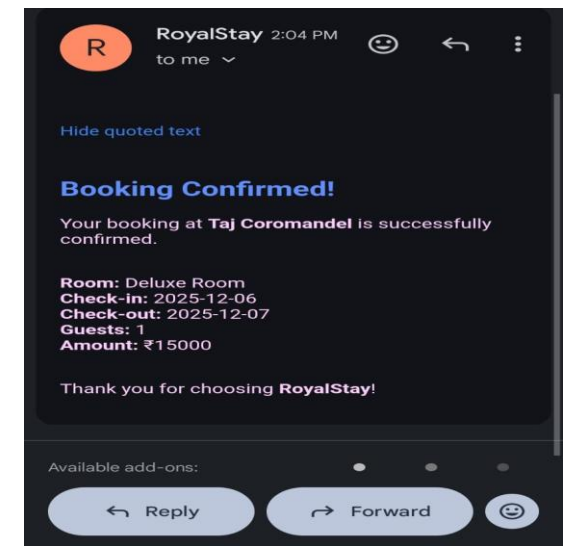
User Portal



Admin Portal



Mail Notification



➤ **Link :** <https://royalstayhotel.great-site.net/user/signin.php>



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Implementation :

```
File Edit View

<?php
session_start();
include "config.php";

// Fetch dynamically added cities (only cities with id > 6)
$newCities = $conn->query("SELECT * FROM cities WHERE id > 6 ORDER BY id ASC");

// Fixed icon cycle for new cities
$iconList = [
    "fa-solid fa-tree-city",
    "fa-solid fa-building",
    "fa-solid fa-mountain-city",
    "fa-solid fa-city",
    "fa-solid fa-map-location-dot"
];
?>
<!DOCTYPE html>
<html>
<head>
<title>RoyalStay - Find Hotels</title>

<link href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.2/dist/css/bootstrap.min.css" rel="stylesheet">
<link href="https://cdnjs.cloudflare.com/ajax/libs/font-awesome/6.5.0/css/all.min.css" rel="stylesheet">

<style>

body {
    background: url('https://images.unsplash.com/photo-1590490360182-c31d21fd1271?auto=format&fit=crop&w=1950&q=80') center/cover no-repeat
    fixed;
    font-family: 'Poppins', sans-serif;
}
```



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Conclusion:

- Online Hotel Reservation System simplifies hotel management.
- Guests can book rooms easily from anywhere.
- Ensures accuracy, speed, and convenience.
- Helps hotels maintain organized booking records.
- Enhances overall guest satisfaction and operational efficiency.



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Future Scope :

- **Dynamic room pricing** can be implemented using machine learning to adjust prices based on demand, season, and user behavior.
- **Mobile application support** can be introduced for easier booking, cancellation, and feedback submission from anywhere.
- **Advanced loyalty & personalization features** like reward points, personalized offers, and targeted notifications can be added to improve guest engagement.
- Providing **automated chatbots** to assist guests with bookings, FAQs, and cancellations.



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References :

- Stringam, B. B., & Gerdes, J. H. (2021). Hotel and guest room technology. University of South Florida (USF) M3 Publishing, 17(9781732127593), 6.
- Arapou, D., & Kapiki, S. (2023). Enhancing the Guest Experience in Hospitality through Technology: An Outlook of the Future.
- Gunjal, P. M. D. S. (2025). Hotel Booking Web Application using MERN Stack.
- Halim, A., Rohman, A., & Ifvochy, J. (2025). The Development of A Hotel Room Reservation Application Using The Laravel Framework at Abadi Hotel. ARRUS Journal of Engineering and Technology, 5(1), 88-100.



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THANK YOU!